

**EARLY DETECTION OF LITHIUM
DENDRITE RISK IN EV BATTERIES
USING QUANTUM SUPPORT VECTOR
MACHINES**

by

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Vishak Senthilkumar

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ABSTRACT

The overall number of electric vehicles (EVs) on the roads is increasing and the need for the safety and reliability of their batteries is becoming more important. Dendrite formation is one of the most critical issues facing an electric vehicle's battery and can cause not only safety hazards but also destruction of the battery itself. This project offers a unique procedure for spotting battery dendrite risk early, using a hybrid machine learning model that merges both classical and quantum computing methods. Quantum Support Vector Machines classify the health of the battery time-series data, which consists of voltage, current and temperature. The sensors that are placed on the battery gather continuous data for a certain period of time and this data is then divided into smaller segments using a sliding window technique of pre-processing. For each segment, various statistics including average, variance and rate of change are computed to identify potential features in the battery's behavior. Features that have been selected are subsequently provided to the hybrid model where the QSVM further pushes the feature transformation process making it easier to detect very slight abnormalities which are signs of dendrite growth. It is the objective of this model to generate real-time, very precise predictions of dendrite risk that will facilitate quick actions taken and thus better management of the overall battery. The fusion of quantum computing and traditional machine learning techniques is what marks a key milestone for the EVs battery health monitoring sector.

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LIST OF ABBREVIATIONS

QSVM Quantum Support Vector Machine

QML Quantum Machine Learning

SVM Support Vector Machine

EV Electric Vehicle

QAOA Quantum Approximate Optimization Algorithm

NISQ Noisy Intermediate-Scale Quantum

CHAPTER 1

INTRODUCTION

EVs can contribute to sustainable transportation and lithium-ion batteries are the key to their operation. The charging process subjects the EV batteries to different electrical and thermal conditions, which when allowed to get out of check may lead to abnormal behavior. Thermal runaway is one of the most important safety issues since the phenomenon, which is defined as an abrupt rise in temperature, may cause battery deterioration or system malfunction. Such abnormal conditions must therefore be identified early in order to be able to ensure that operations are safe and also to guarantee reliability of the system.

Due to the growing implementation of sensor-based battery management systems, nowadays it is possible to collect a considerable amount of time-stamped battery charging data in real time. This data is usually continuous measurements, including voltage, current, temperature and state of charge, measured during the charge, which are recorded during the charging cycle. This data is usually analyzed with machine learning methods and used to identify abnormal charging conduct. Nevertheless, most of the current methods are based on classical machine learning models only, and these might not be able to depict the complicated nonlinear relationships that can exist in multivariate time-series sensor data.

The new developments in quantum computing provided quantum-inspired machine learning (QML) methods that present alternative data representation and transformation techniques. Despite the fact that mod-

ern quantum devices have not offered any computational advantage of practical datasets, hybrid learning systems with classical and quantum-inspired components have been considered, because the feature representation can be improved. Here, it is assumed that QML will not replace the classical machine learning, but it represents another processing layer that can be added to the already existing learning pipeline.

This paper suggests a hybrid classical-quantum learning model to detect anomalies in EV battery charging systems with real time stamped sensor data. It is based on the suggested procedure that combines the classical time-series preprocessing and feature extraction with a quantum-inspired processing layer that is aimed at executing nonlinear feature transformation. The resulting hybrid feature representation is then studied with a classical classifier to determine abnormal charging patterns that relate to thermal runaway.

This project will have the main goal of designing and deploying an entire learning pipeline showing the extent to which advanced machine learning techniques can be integrated in a safety-critical and real world application. Instead of prioritizing the superior performance or computational acceleration, it focuses on the architectural design, feasibility, and the systematization of evaluation of hybrid feature representations of time-series anomaly detection. The efficiency of the suggested framework is assessed with the help of the data set of time-stamped sensor readings that were recorded in EV battery charge and thermal runaway events.

CHAPTER 2

LITERATURE REVIEW

Table 2.1 Literature Review

Paper	Methodology	Limitations	Ideas for Adoption
Comparative Investigation of Quantum and Classical Kernel Functions Applied in Support Vector Machine Algorithms	Performs a comparative study of classical kernels and quantum kernels within the SVM framework. Different quantum feature maps are evaluated using QSVM on benchmark datasets.	Uses quantum simulators; high kernel computation cost; limited scalability; lacks real-world application focus.	Adopt the quantum kernel comparison framework to evaluate classical SVM versus QSVM for battery time-series features, justifying improved separability for early dendrite-risk patterns.

Paper	Methodology	Limitations	Ideas for Adoption
Assessment of Quantum Feature Map for Binary Forest Classification using QSVM	Evaluates multiple Pauli-based quantum feature maps for binary classification. Features are normalized, reduced, encoded, and classified using QSVM with Pareto-front analysis.	Relies on simulators; inconsistent QSVM superiority; scalability limitations in NISQ-era hardware.	Apply quantum feature map selection strategies to battery sensor data and use Pareto analysis to balance accuracy and computational cost.
Hybrid Quantum–Classical Framework for Urban Traffic State Classification	Implements a hybrid pipeline combining classical feature extraction with quantum encoding and QSVM-based classification.	Limited dataset; simulator-based execution; real-time scalability not addressed.	Use the hybrid classical–quantum architecture for battery monitoring by applying QSVM after classical preprocessing of voltage and temperature data.

Paper	Methodology	Limitations	Ideas for Adoption
AutoQML: A Framework for Automated Quantum Machine Learning	Introduces automated quantum model selection, feature encoding, and hyperparameter tuning in hybrid workflows.	High computational overhead; simulator dependence; not tailored for time-series data.	Adopt automated quantum kernel and hyperparameter tuning to reduce manual QSVM optimization for battery datasets.
Real-Time Quantum Machine Learning-Based Anomaly Detection for Lithium-Ion Battery Packs	Proposes quantum-enhanced K-Means clustering for real-time anomaly detection using battery sensor data.	Focuses on clustering; requires multiple qubits; real-time hardware feasibility unclear.	Extend the anomaly detection framework by replacing clustering with QSVM-based supervised dendrite-risk classification.
Quantum Kernel-Based Support Vector Machine with QAOA Embedding for Lung Cancer Prediction	Uses QAOA-based quantum feature embedding with QSVM, achieving better results for small datasets.	Simulator execution; high cost; evaluated only on static medical data.	Adopt QAOA-based quantum embedding for capturing nonlinear patterns in reduced battery feature sets.

CHAPTER 3

SYSTEM DESIGN

3.1 DESIGN OVERVIEW

The proposed system design aims at developing a dependable hybrid Quantum Support Vector Machine (QSVM)-based system for detecting anomalies and assessing risks in electric vehicle (EV) battery charging systems. The architecture is inspired by existing quantum machine learning-based battery anomaly detection solutions, particularly those based on unsupervised clustering algorithms such as K-means.

Although clustering-based methods can identify coarse deviations in battery behaviour, they often fail to detect subtle early-stage anomalies and lack clear decision margins. To overcome these limitations, the proposed system replaces clustering with a margin-based hybrid QSVM classifier combined with a risk-oriented decision layer.

The system follows a modular pipeline architecture in which each module performs a specific transformation and passes its output to the next stage. This design improves interpretability, scalability, and ease of integration with existing battery management systems.

3.2 SYSTEM OVERVIEW

The proposed EV battery monitoring framework follows a sequential processing pipeline that transforms raw sensor measurements into battery risk predictions.

The overall workflow of the system is illustrated in Figure ??.

The process begins with acquiring battery sensor measurements and proceeds through preprocessing, feature engineering, quantum feature transformation, and QSVM-based classification.

The major stages of the workflow are summarized as follows:

1. **Data Acquisition:** Battery sensor data such as voltage, current, temperature, and state of charge are collected during charging and discharging cycles.
2. **Data Preprocessing:** Raw measurements are filtered to remove noise, normalized to maintain consistent scales, and segmented using sliding windows to capture short-term battery behaviour.
3. **Feature Engineering:** Statistical and electrochemical features are extracted from each window to represent battery behaviour in a structured feature space.
4. **Quantum Feature Mapping:** Classical feature vectors are mapped into a higher-dimensional Hilbert space using quantum-inspired encoding techniques.
5. **QSVM Classification:** A hybrid Quantum Support Vector Machine classifier learns decision boundaries between different battery risk levels.
6. **Risk Assessment:** The predicted QSVM output is interpreted and mapped to battery risk categories, producing monitoring reports and confidence scores.

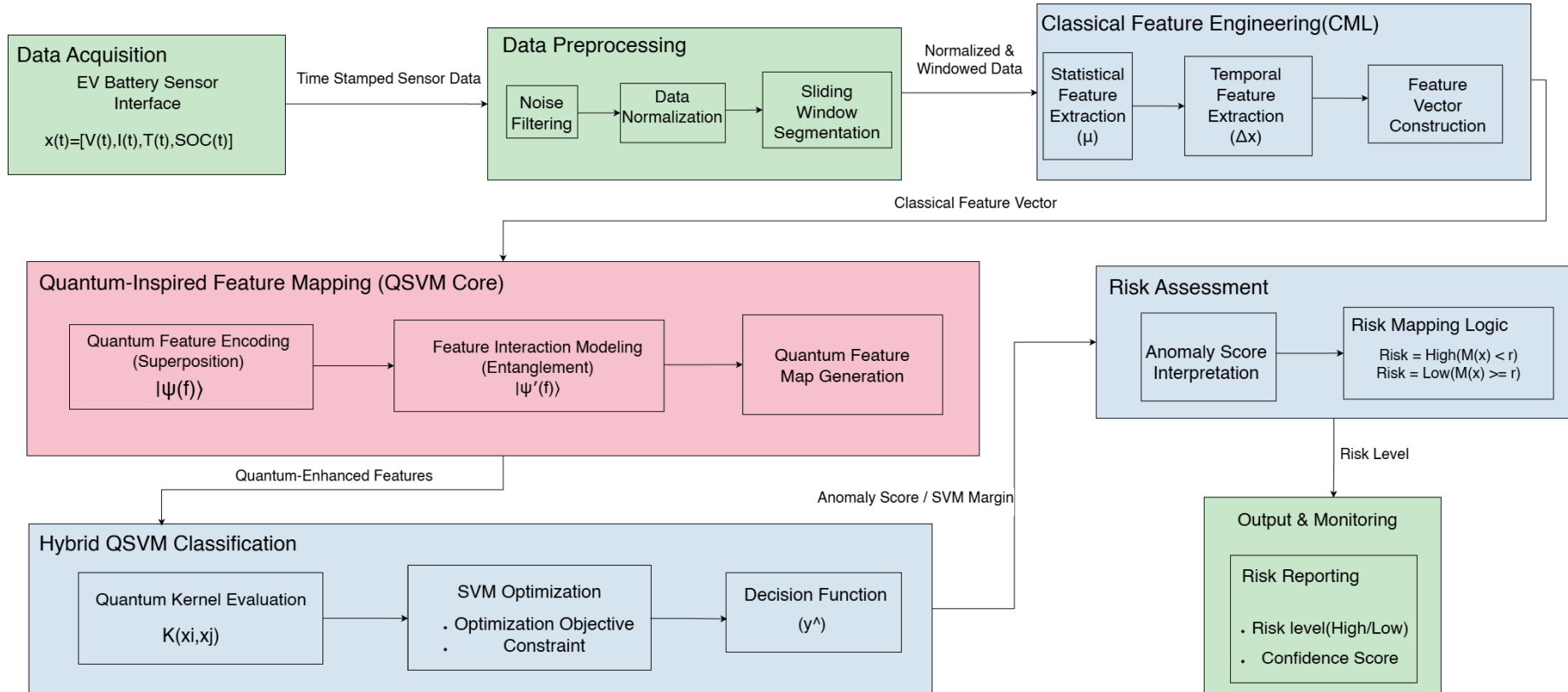


Figure 3.1 Architecture of the Proposed QSVM-Based EV Battery System

3.2.1 Data Acquisition Module

The data acquisition module gathers time-stamped sensor data from EV battery packs during charging cycles. Each record contains synchronized measurements of:

- Voltage
- Current
- Temperature
- State of Charge (SOC)

These parameters collectively form a multivariate time-series representation of battery behaviour. Since battery anomalies often result from interactions between multiple parameters rather than single threshold violations, collecting multiple sensor signals provides a more comprehensive monitoring framework.

Algorithm 1 Battery Data Loading Module

Input: NASA battery .mat files

Output: Structured dataset D

for each battery file B_i **do**

Load B_i using MATLAB reader

Extract cycle structure

Separate discharge, charge, impedance cycles

Store structured data in D

end for

return D

3.2.2 Data Preprocessing Module

Raw sensor data may contain noise, outliers, and variations in measurement scales. Therefore, preprocessing is necessary before feature extraction.

The preprocessing module performs:

- Noise filtering

- Data normalization
- Sliding window segmentation

Noise filtering reduces sensor disturbances, normalization ensures comparable feature ranges, and sliding window segmentation divides the time-series data into fixed-length windows that represent short-term charging behaviour.

Algorithm 2 Sliding Window Degradation Modeling

Input: Feature matrix F

Output: Windowed dataset W

Define window size w and stride s

for each battery b **do**

for each window segment **do**

 Compute capacity trend (slope)

 Compute capacity change (last - first)

 Compute impedance trend

 Compute degradation variance

 Store window features

end for

end for

return W

3.2.3 Classical Feature Engineering Module

After preprocessing, statistical feature engineering is applied to extract meaningful descriptors from each window.

Typical features include:

- Mean
- Variance
- Rate of change
- Stability indicators

These features form a structured classical feature vector representing the battery behaviour within a time window. This stage reduces the

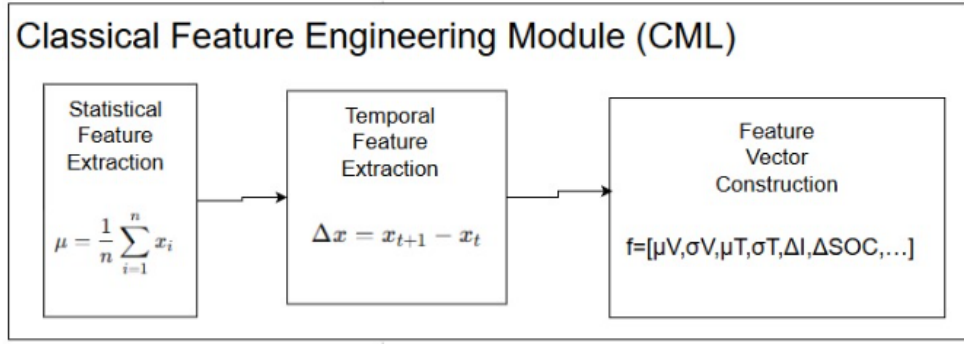


Figure 3.2 Classical Feature Engineering Module

dimensionality of raw sensor data and provides interpretable inputs for further analysis.

Algorithm 3 Statistical and Electrochemical Feature Extraction

Input: Dataset D

Output: Feature matrix F

for each discharge cycle c in D **do**

 Extract capacity

 Compute voltage statistics (mean, std, min, max, range)

 Compute current statistics (mean, std)

 Compute temperature statistics (mean, std, min, max)

 Extract impedance parameters (Re, Rct)

end for

Compute Coulombic Efficiency:

$$CE = \frac{Q_{discharge}}{Q_{charge}}$$

Merge all features into F

return F

As illustrated in Figure 3.2, the classical feature engineering stage extracts statistical and temporal features from battery sensor data.

3.2.4 Quantum-Inspired Feature Mapping Module

The quantum-inspired feature mapping module converts classical feature vectors into a higher-dimensional Hilbert space using quantum encoding techniques.

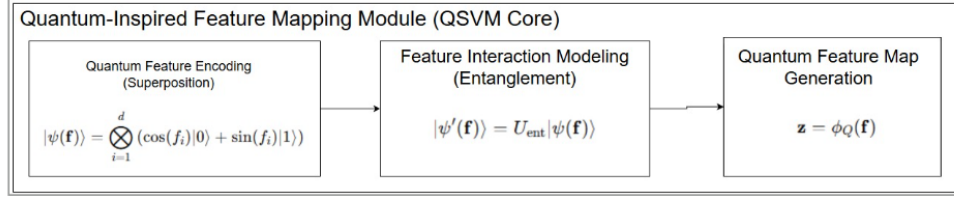


Figure 3.3 Quantum-Inspired Feature Mapping Module

Initially, classical features are encoded using superposition-based representations, allowing multiple components to be represented simultaneously. Entanglement-inspired transformations are then applied to capture interactions between features.

These transformations increase the expressive capability of the feature space without requiring physical quantum hardware.

3.2.5 Hybrid QSVM Classification Module

The proposed system uses a hybrid QSVM classifier to detect anomalous charging behaviour.

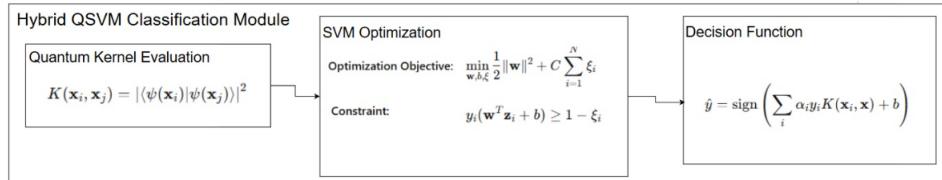


Figure 3.4 Hybrid Quantum Support Vector Machine Classification Module

Unlike clustering-based approaches, QSVM learns a margin-based decision boundary that separates different battery degradation patterns in the transformed feature space.

The similarity between feature vectors is computed using a quantum kernel:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

The QSVM combines quantum kernel feature mapping with classical SVM optimization.

3.2.6 Risk Assessment and Output Generation

The system includes a risk assessment module that interprets the QSVM output to classify battery charging behaviour into different risk levels.

The classification output generated by the QSVM model is mapped to predefined battery health categories based on degradation patterns.

The battery state is categorized into the following risk levels:

- Very Low Risk
- Low Risk
- Medium Risk
- High Risk
- Critical Risk

These categories represent increasing levels of battery degradation and potential dendrite formation.

The final system outputs include:

- Predicted risk level
- Confidence score
- Monitoring reports

3.3 DESIGN SUMMARY

The proposed system extends existing battery anomaly detection frameworks by incorporating quantum-inspired feature mapping within a classical machine learning pipeline. The use of a hybrid QSVM classifier enables better separation between normal and abnormal bat-

Table 3.1 Battery Risk Level Categories

Class Label	Risk Level
0	Very Low Risk
1	Low Risk
2	Medium Risk
3	High Risk
4	Critical Risk

tery behaviour while maintaining interpretability and computational efficiency.

CHAPTER 4

Implementation and Results

4.1 OVERVIEW OF THE PROPOSED ARCHITECTURE

The proposed hybrid Quantum Machine Learning framework was implemented using a modular pipeline that combines classical signal processing with quantum-enhanced machine learning techniques.

The overall workflow of the system includes the following stages:

1. Environment Setup
2. Data Loading
3. Timestamp Alignment
4. Missing Value Handling
5. Statistical Feature Engineering
6. Dendrite-Oriented Feature Engineering
7. Sliding Window Modelling
8. Window-Level Feature Construction
9. Quantum Feature Mapping
10. Quantum Support Vector Machine Classification
11. Risk Assessment and Reporting

Each module performs a specific transformation on the battery sensor data before passing it to the next stage of the pipeline.

4.2 IMPLEMENTATION ENVIRONMENT

The implementation was carried out using the Python programming language with several scientific computing and quantum computing li-

braries.

Table 4.1 Software Environment Used for Implementation

Tool / Library	Purpose
Python	Core programming language
NumPy	Numerical computation
Pandas	Data processing
Scikit-learn	Classical machine learning utilities
Matplotlib / Seaborn	Data visualization
Qiskit	Quantum circuit simulation
Qiskit Machine Learning	Quantum kernel and QSVM implementation
Google Colab	Development and execution environment

The implementation was executed using the Google Colab platform, which allows integration of Python libraries and cloud-based computational resources.

4.3 DATASET DESCRIPTION

The NASA Lithium-Ion Battery Dataset was used for this study. The dataset consists of battery cycle measurements collected during laboratory experiments.

Each battery file contains multiple operational cycles including:

- Charge cycles
- Discharge cycles
- Impedance measurements

Battery measurements include the following parameters:

- Voltage
- Current
- Temperature
- Capacity
- Time

Table 4.2 Dataset Summary

Parameter	Value
Total Battery Files	35
Valid Batteries Processed	33
Total Cycles Loaded	7313
Discharge Cycles Used	4600
Valid Cycles after Filtering	2667

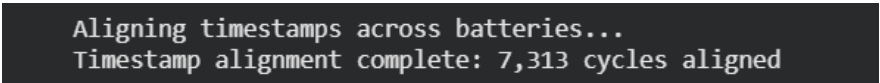
4.4 DATA LOADING

Battery data stored in MATLAB (.mat) format was loaded using the `scipy.io` library. The MATLAB structures were parsed to extract discharge cycle measurements.

Only discharge cycles were used because they contain the relevant electrochemical behavior needed to analyze battery degradation patterns.

4.5 TIMESTAMP ALIGNMENT

Battery cycles contain time arrays representing measurements recorded during each cycle. These timestamps were aligned with cycle indices to maintain consistency across multiple battery datasets.



```
Aligning timestamps across batteries...
Timestamp alignment complete: 7,313 cycles aligned
```

Figure 4.1 Timestamp Alignment Across Battery Cycles

Timestamp alignment enables proper comparison of degradation patterns across batteries.

4.6 MISSING VALUE HANDLING

Different cycle types contain different measurement fields. Therefore, missing values were handled using safe extraction techniques.

```

... Extracting raw data with missing value handling...
Raw data extraction complete: 4,600 cycles

Missing value summary:
capacity: 2692 missing
voltage: 0 missing
current: 0 missing
temperature: 0 missing
time: 0 missing

```

Figure 4.2 Handling Missing Values in Battery Dataset

Median-based imputation was applied to fill missing entries while preserving statistical characteristics of the dataset.

4.7 STATISTICAL FEATURE ENGINEERING

Time-series battery signals were transformed into statistical descriptors to provide structured inputs for machine learning models.

The following statistical features were extracted:

- Mean
- Standard deviation
- Minimum value
- Maximum value
- Median

Electrochemical impedance parameters such as resistance components were also extracted to capture degradation behavior.

4.8 DENDRITE-ORIENTED FEATURE ENGINEERING

Additional electrochemical indicators were extracted to detect early signs of dendrite formation.

One important feature used is the differential capacity curve defined as

$$\frac{dQ}{dV} \quad (4.1)$$

Changes in the differential capacity curve can indicate structural degradation within the battery.

Additional descriptors extracted include:

- Skewness
- Kurtosis
- Entropy
- Interquartile Range

These features capture subtle variations in battery behavior that may indicate early dendrite growth.

4.9 SLIDING WINDOW MODELLING

Battery degradation is a gradual process. To capture temporal degradation patterns, sliding window segmentation was applied across battery cycles.

Table 4.3 Sliding Window Parameters

Parameter	Value
Window Sizes	5, 10, 20 cycles
Stride	6 cycles

Within each window, degradation trends such as capacity slope and degradation variance were computed.

4.10 WINDOW-LEVEL FEATURE CONSTRUCTION

For each sliding window segment, several degradation trend features were extracted.

These window-level features represent temporal degradation patterns observed in battery cycles.

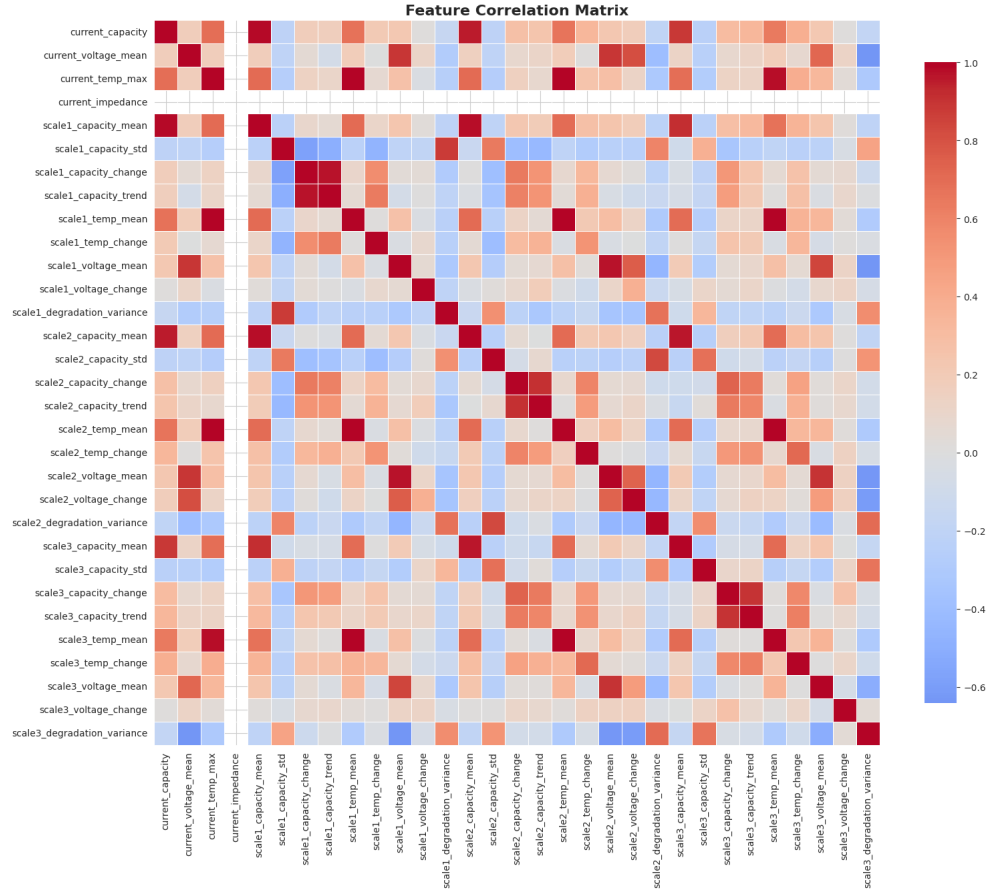


Figure 4.3 Feature Extraction from Sliding Window Segments

4.11 QUANTUM FEATURE MAPPING

Principal Component Analysis (PCA) was used to reduce the dimensionality of the classical feature space before quantum encoding.

The original feature space was reduced to four principal components. These components were normalized to the interval $[0, \pi]$ for quantum circuit rotations.

Quantum feature encoding was then applied using parameterized quantum gates and entanglement operations.

The quantum circuit used for kernel generation in the QSVM model is shown in Figure 4.4.

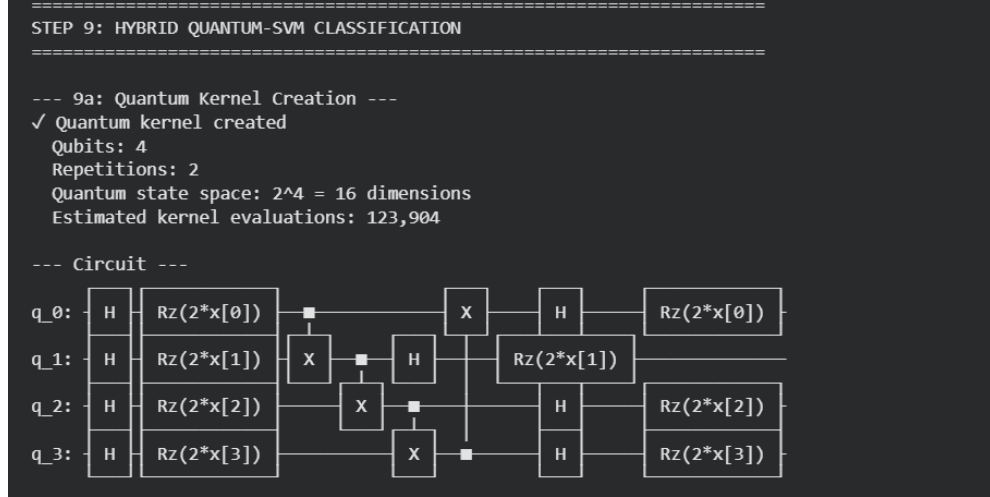


Figure 4.4 Quantum Kernel Circuit Used for QSVM Feature Encoding

4.12 QUANTUM KERNEL SUPPORT VECTOR MACHINE

The classification stage uses a Quantum Kernel Support Vector Machine (QSVM).

The quantum feature map transforms classical feature vectors into quantum states as follows:

$$|\psi(x)\rangle = U_\phi(x)|0\rangle \quad (4.2)$$

The similarity between two samples is computed using the quantum kernel function:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2 \quad (4.3)$$

The resulting quantum kernel matrix is used by a classical SVM classifier to separate battery risk categories.

4.13 TRAIN-TEST SPLIT

The dataset was divided into training and testing sets to evaluate the model performance.

Table 4.4 Dataset Split

Dataset Portion	Percentage
Training Set	80%
Testing Set	20%

4.14 TEST CASES

Several test scenarios were created to validate the ability of the model to detect different battery degradation states.

Table 4.5 Test Case Examples

Test Case	Input Condition	Expected Risk Level
TC1	Stable voltage and temperature	Very Low Risk
TC2	Slight voltage fluctuation	Low Risk
TC3	Moderate degradation trend	Medium Risk
TC4	Rapid capacity decline	High Risk
TC5	Severe degradation indicators	Critical Risk

4.15 RESULTS AND DISCUSSION

The hybrid quantum-classical framework successfully identified degradation patterns within the battery dataset. The quantum feature transformation improved the separability of battery health states in the feature space.

The predicted risk distribution and confidence levels obtained from the QSVM model are shown in Figure 4.5.

4.16 PERFORMANCE METRICS

To evaluate the effectiveness of the proposed system, several performance metrics were used.

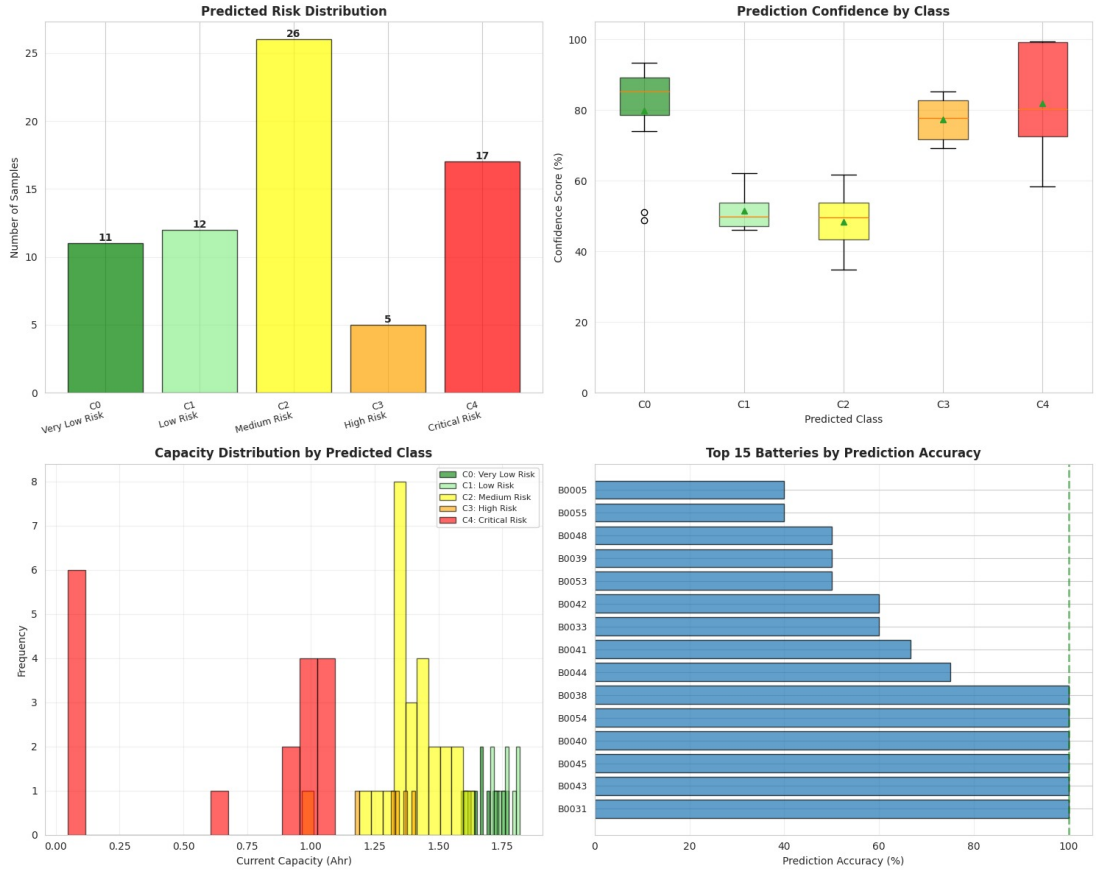


Figure 4.5 Predicted Risk Distribution, Prediction Confidence and Battery Accuracy Analysis

4.16.1 Accuracy

Accuracy measures the proportion of correctly classified samples among all test instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.4)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively.

4.16.2 Recall

Recall evaluates the model's ability to correctly identify actual anomalous battery conditions.

$$Recall = \frac{TP}{TP + FN} \quad (4.5)$$

High recall is critical in EV battery monitoring because missing an anomaly may lead to severe safety risks.

4.16.3 Feature Space Separability

Feature Space Separability (FSS) measures the separation between normal and anomalous samples after quantum feature transformation.

$$FSS = \frac{\|\mu_{normal} - \mu_{anomaly}\|_2}{\sigma_{normal} + \sigma_{anomaly}} \quad (4.6)$$

Higher values of FSS indicate better class separability.

4.16.4 Variance Amplification Ratio

Variance Amplification Ratio (VAR) quantifies the increase in feature variance produced by the quantum transformation.

$$VAR = \frac{Var(Q(x))}{Var(x)} \quad (4.7)$$

where x represents the classical feature vector and $Q(x)$ represents the quantum-transformed feature vector.

Figure 4.6 presents the confusion matrix and per-class F1 score of the QSVM classifier.

The results demonstrate that the hybrid quantum-classical learning framework improves the representation of battery degradation patterns and enables early detection of lithium dendrite risk.

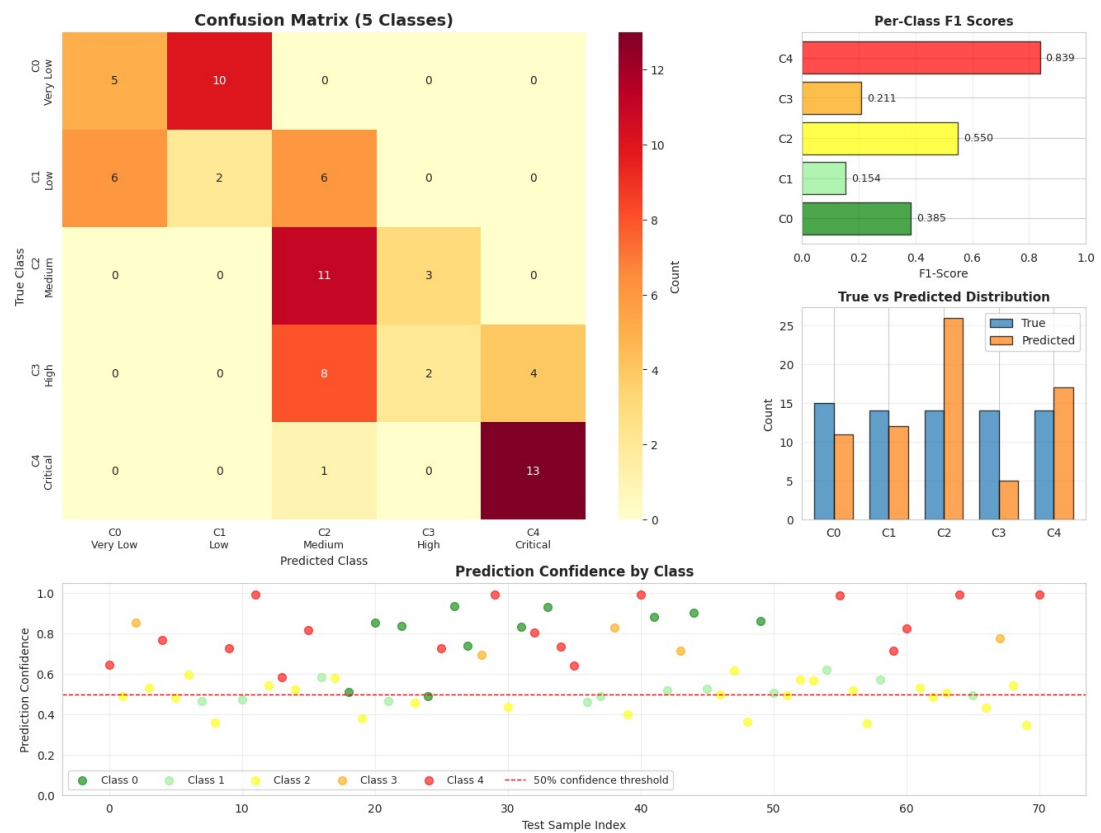


Figure 4.6 Confusion Matrix and Per-Class F1 Score of the QSVM Model

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 SUMMARY OF THE WORK

This powerful data processing pipeline established a cornerstone. We successfully converted raw, heterogeneous battery data into a standardized, clean, and feature-rich dataset. These multi-scale temporal features now prepare for the next phase of this project: the development and training of Quantum Machine Learning models for predictive analytics, with the main focus on early detection and understanding of battery dendrite formation to enhance battery safety, life, and performance. Feature vectors generated at this step are the required input for advanced machine learning algorithms and allow the study of subtle signatures of degradation that might indicate dendrite growth.

5.2 QUANTUM MACHINE LEARNING ENHANCEMENTS

Future improvements to the system can enhance quantum feature representation and classification performance.

5.2.1 Expand Qubit Count

Increasing the number of qubits from 4 to 6–8 would expand the feature space to 64–256 dimensions.

Maintaining PCA variance between 80–90% may improve kernel expressiveness and reduce support vector ratios.

5.2.2 Advanced Quantum Feature Maps

Future implementations may replace the existing feature map with advanced circuits such as:

- Pauli Feature Maps
- IQP circuits

Trainable quantum kernels can also be introduced to maximize class separability.

5.2.3 Hybrid Variational Quantum Circuits

Variational Quantum Eigensolver (VQE) or Quantum Neural Networks may be integrated before the SVM classifier to learn latent quantum representations.

5.3 CLASSICAL MACHINE LEARNING ENHANCEMENTS

5.3.1 Improved Label Design

Future work may incorporate physics-based thresholds such as State of Health (SOH) limits instead of statistical quantile labels.

5.3.2 Enhanced Feature Engineering

Additional electrochemical features such as Electrochemical Impedance Spectroscopy (EIS) parameters may be included.

Wavelet transforms can also be used to extract multi-frequency degradation signatures.

5.3.3 Classical Baseline Comparison

Benchmarking against classical models such as:

- Random Forest
- XGBoost
- LSTM

can help evaluate the practical advantages of quantum models.

5.4 DATA AND SCALABILITY

Future experiments may include larger datasets from NASA and CALCE battery repositories to improve generalization.

Testing on real quantum hardware using IBM Quantum devices can also evaluate practical quantum advantage.

5.5 SYSTEM INTEGRATION AND DEPLOYMENT

The risk prediction module can be integrated into Battery Management Systems (BMS) as a REST API microservice.

Real-time dashboards can visualize battery risk levels and trigger alerts when critical thresholds are reached.

Explainability methods such as SHAP can be used to interpret feature contributions in the model predictions.

5.6 MULTI-CHEMISTRY EXTENSION

Future systems may extend the methodology to other battery chemistries such as lithium-sulfur or solid-state batteries, enabling broader battery safety monitoring applications.